

# Adnan Armouti

New York, NY | [aa2546@cornell.edu](mailto:aa2546@cornell.edu) | [adnan-armouti.github.io](https://adnan-armouti.github.io) | [Google Scholar](#) | [GitHub](#)

**Interests:** Physical AI, RF Spatial Intelligence, Differentiable Rendering, Inverse Problems, Radar

## Education

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| <b>Ph.D. Candidate in Computer Science</b> , Cornell University | Aug 2026–Present  |
| <i>Advisor:</i> Rajalakshmi Nandakumar                          |                   |
| <b>M.S., Computer Science</b> , Cornell University              | Aug 2023–May 2026 |
| <b>M.S., Electrical and Computer Engineering</b> , UCLA         | Sep 2021–Jun 2023 |
| <b>B.S., Electrical Engineering</b> , UCLA                      | Sep 2017–Jun 2021 |
| <b>B.A., Economics</b> , UCLA                                   | Sep 2017–Jun 2021 |

## Current Research

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**Towards RF Spatial Intelligence: Physics-Based Differentiable Rendering for Inverse Problems in Millimeter-Wave Radar**

*Committee:* Rajalakshmi Nandakumar (Chair), Andrew Owens, Steve Marschner, Alyosha Molnar

- I develop explicit, physics-based differentiable renderers for mmWave radar that fit real captures via material and surface normal estimation [1, 2].
- Explicit modeling of independent scene and sensor representations enables scene- and sensor-level editing to solve inverse problems such as 2D-to-3D lifting [2], super-resolution [2], and novel view synthesis [1].

## Publications

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### Selected Papers

\* equal contribution

- [1] [Adnan Armouti](#), Yixuan Gao, Rajalakshmi Nandakumar. “3D Point Splatting for mmWave Radar Novel View Synthesis.” *NeurIPS 2026*.
  - [2] [Adnan Armouti](#), Yixuan Gao, Rajalakshmi Nandakumar. “mmIR: Frequency-Space Inverse Rendering for 3D Millimeter-Wave Radar ADC Synthesis.” *ECCV 2026*.
  - [3] Alexander Vilesov\*, Pradyumna Chari\*, [Adnan Armouti](#)\*, Anirudh Bindiganavale Harish, Kimmaya Kulkarni, Ananya Deoghare, Laleh Jalilian, Achuta Kadambi. “Blending Camera and 77 GHz Radar Sensing for Equitable, Robust Plethysmography.” *ACM Transactions on Graphics (SIGGRAPH 2022)*.
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- [4] Tanvir Ahmed, Yixuan Gao, [Adnan Armouti](#), Rajalakshmi Nandakumar. “mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption.” *MobiCom 2026*.
  - [5] Pradyumna Chari\*, Anirudh Bindiganavale Harish\*, [Adnan Armouti](#), Alexander Vilesov, Sanjit Sarda, Laleh Jalilian, Achuta Kadambi. “Implicit Neural Models to Extract Heart Rate from Video.” *ECCV 2024*.
  - [6] Kai Del Regno, Alexander Vilesov, [Adnan Armouti](#), Anirudh Bindiganavale Harish, Selim Emir Can, Ashley Kita, Achuta Kadambi. “Thermal Imaging and Radar for Remote Sleep Monitoring of Breathing and Apnea.” *arXiv preprint (2024)*.

## Patents

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- [US1] Achuta Kadambi, Laleh Jalilian, Pradyumna Chari, Chinmay Talegaonkar, Doruk Karınca, Maxime Cannesson, Krish Kabra, Omid Salehi-Abari, Ashley Kita, [Adnan Armouti](#). “[Systems and Methods for Measuring Vital Signs Using Multimodal Health Sensing Platforms](#).” U.S. Patent Application 2023/0233091, 2023.
- [US2] Achuta Kadambi, Ashley Kita, Alexander Vilesov, Kai Del Regno, Selim Emir Can, Laleh Jalilian, Anirudh Bindiganavale Harish, [Adnan Armouti](#). “Methods and Apparatus to Detect and Classify Forms of Sleep Apnea.” U.S. Provisional Application, 2024.

## Honors and Awards

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|                                                       |            |
|-------------------------------------------------------|------------|
| NVIDIA Academic Grant                                 | Jun 2026   |
| UCLA ECE Outstanding Master of Science Student Award  | Jun 2023   |
| UCLA ECE Distinguished Master’s Thesis Research Award | May 2023   |
| UCLA ECE VMG GSR Scholarship                          | Jan 2022   |
| NSF REU Fellowship                                    | Jun 2021   |
| Intel SRC URP Scholarship                             | Jun 2020   |
| UCLA ECE Dean’s Honors List                           | 2020, 2021 |

## Research Experience

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**Wireless Systems Lab**, Cornell University, New York, NY Aug 2023–Present  
*PI*: Rajalakshmi Nandakumar

- Built mmIR, an open-source differentiable FMCW radar inverse renderer with multi-bounce propagation, polarization, and diffraction; 0.914 RA correlation on 13 scenes vs. Sionna-RT’s 0.307 [2].
- Built 3DPS, the first differentiable point renderer for radar novel view synthesis, producing ADC, range profiles, and range–azimuth maps from one model;  $1.7\text{--}5.2\times$  over NVS baselines [1].
- Contributed to mmFHE, the first mmWave sensing pipeline (signal processing and ML inference) run end to end under fully homomorphic encryption [4].

**Visual Machines Group**, UCLA Sep 2020–Jun 2023  
*PI*: Achuta Kadambi

- Led a successful \$1M DARPA grant proposal for contactless, equitable health sensing.
- Co-led camera–radar fusion for skin-tone-equitable remote plethysmography [3].
- Co-implemented a neural implicit decomposition for heart-rate estimation on out-of-distribution subjects [5], and contributed to thermal and radar sleep-apnea monitoring [6].

## Public Demonstrations

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Tanvir Ahmed, Yixuan Gao, [Adnan Armouti](#), Rajalakshmi Nandakumar. “[mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption](#).” *MobiCom Demos 2026* [4].

## Media Coverage

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|-----------------------------------|----------|
| <a href="#">Forbes</a> [3]        | Jan 2023 |
| <a href="#">Daily Bruin</a> [3]   | Sep 2022 |
| <a href="#">UCLA Newsroom</a> [3] | Aug 2022 |

## Open-Source Contributions

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**3DPS** (coming soon). Differentiable point splatting for radar novel view synthesis [1].

**mmIR** ([project page](#)). Differentiable FMCW radar inverse renderer [2].

**EquiPleth** ([GitHub](#)). Camera–radar fusion for equitable plethysmography [3].

**FastImplicitPleth** ([GitHub](#)). Implicit neural models for heart rate from video [5].

**NonContactApneaDetection** ([GitHub](#)). Thermal and radar sleep-apnea detection [6].

## Theses

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**Fusing Light and Radio Signals for More Accurate and Equitable Blood Flow Estimation**

M.S. Thesis, UCLA

2023

*Committee:* Achuta Kadambi (Chair), Jonathan Kao, Cho-Jui Hsieh, Bolei Zhou

## Teaching

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**Teaching Assistant**, Cornell Tech, Cornell University Aug 2023–Present

- CS 5785 “Applied Machine Learning,” Prof. Kyra Gan (Fall 2023, Fall 2024, Fall 2025, Fall 2026)
- CS 5788 “Introduction to Generative Models,” Prof. Andrew Owens (Spring 2026)
- INFO 5368 “Practical Applications in Machine Learning,” Prof. Angelique Taylor (Spring 2025, Spring 2026)

## Mentorship

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Guanyue (Thomas) Qian, *B.S. Student, NYU*

Jun 2026–Present

Kai Del Regno, *M.S. Student, UCLA*

Sep 2022–Jun 2023

Jianchong (Devin) Ma, *M.S. Student, Stanford University*

Sep 2022–Jun 2023

Zoe (Rui) Ma, *M.S. Student, Columbia University*

Sep 2022–Jun 2023

## Referee Service

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NeurIPS

2026–

## Skills

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- **AI / ML / Computer Vision:** PyTorch, JAX, TensorFlow
- **Computer Graphics:** Mitsuba 3, Dr.Jit, Slang, CUDA
- **Programming & Tools:** Python, C/C++, CUDA, MATLAB; Git, Linux, Docker, L<sup>A</sup>T<sub>E</sub>X

## References

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**Prof. Rajalakshmi Nandakumar**, Cornell University, [rn283@cornell.edu](mailto:rn283@cornell.edu)

**Prof. Achuta Kadambi**, UCLA, [achuta@ee.ucla.edu](mailto:achuta@ee.ucla.edu)

**Prof. Kyra Gan**, Cornell University, [jg2424@cornell.edu](mailto:jg2424@cornell.edu)